

Improved upper bounds on the stabilizer rank of magic states

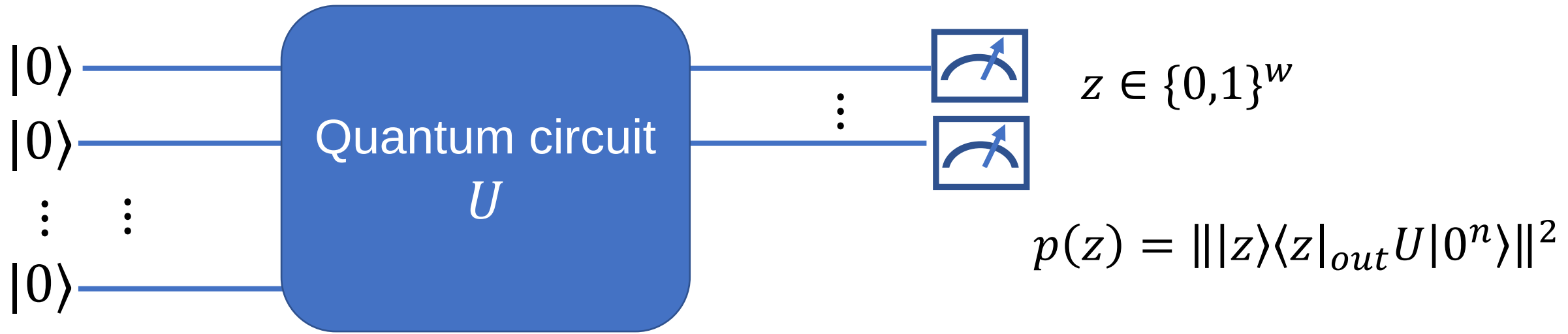
arXiv: 2106.07740

Hammam Qassim, Hakop Pashayan, **David Gosset**

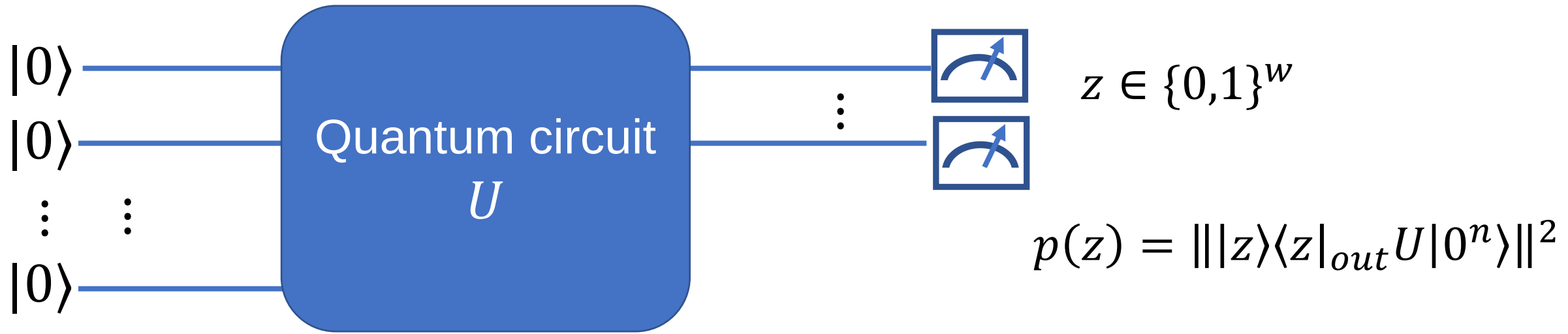
Builds on: Garcia-Ramirez Ph.D thesis 2014
1506.01396 [Bravyi Smith Smolin 15]
1601.07601 [Bravyi DG 16]

....

Classical simulation of quantum circuits

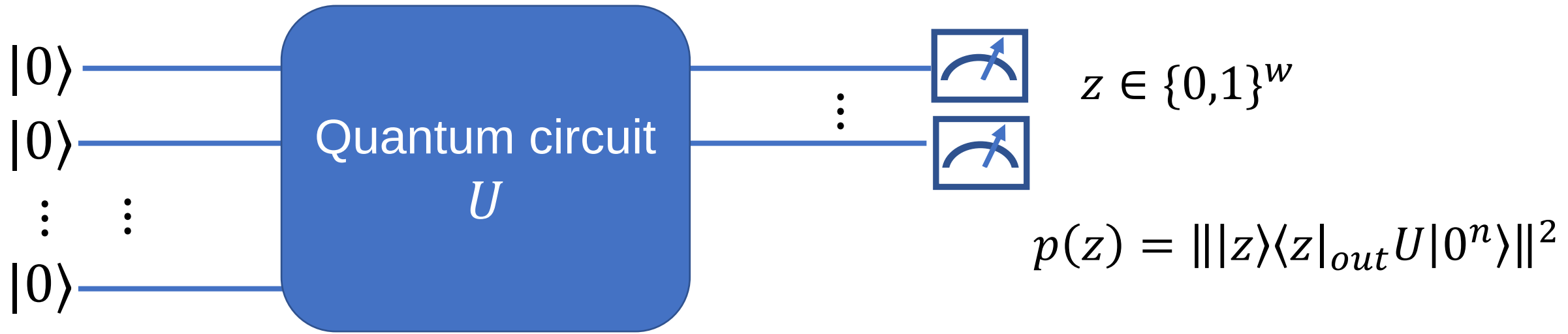


Classical simulation of quantum circuits



Strong simulation: Given x , compute $p(x)$.
(#P-hard. Quantum computers can't do this)

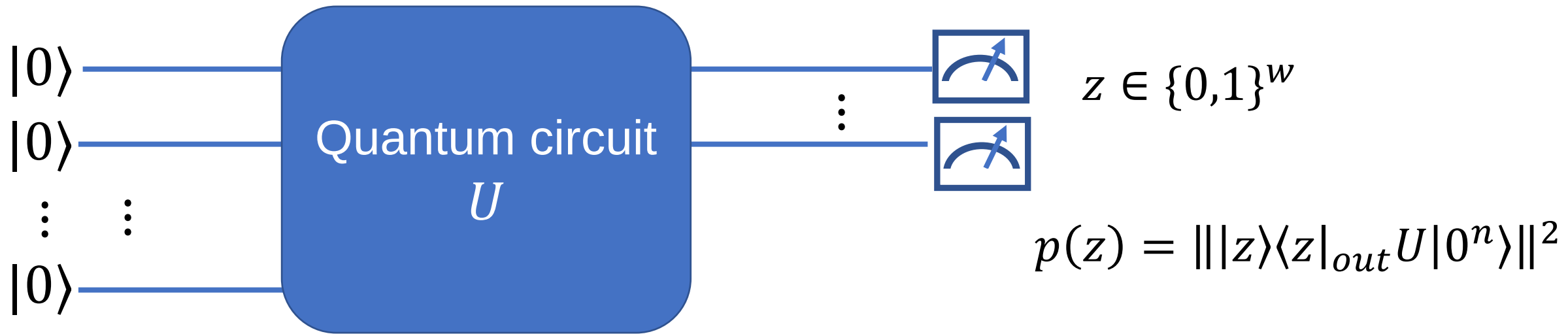
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Weak simulation: Sample a bit string from the distribution p .
(Quantum computers do this)

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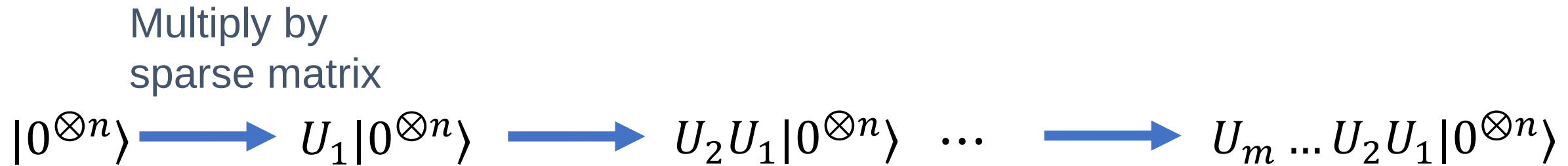
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ϵ -Strong simulation: Given x , compute $\tilde{p}$ such that $(1 - \epsilon)p(x) \leq \tilde{p} \leq (1 + \epsilon)p(x)$.
(#P-hard. Quantum computers can't do this)

Weak simulation: Sample a bit string from the distribution p .
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Classical Simulation

Most algorithms have exponential scaling in the number of qubits or number of gates



Complex vector
 2^n entries

Runtime: $2^n m$

Memory: 2^n

Classical Simulation

Most algorithms have exponential scaling in the number of qubits or number of gates

$$\langle x | U_m \dots U_2 U_1 | 0^{\otimes n} \rangle = \sum_{z_1, z_2 \dots z_{m-1}} \langle x | U_m | z_{m-1} \rangle \dots \langle z_2 | U_2 | z_1 \rangle \langle z_1 | U_1 | 0 \rangle$$

Runtime: 4^m

Memory: $m + n$

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Recursive variant
[Aaronson Chen 2016]

Runtime: $n(2d)^{n+1}$

Memory: $n \log(d)$

Circuit depth



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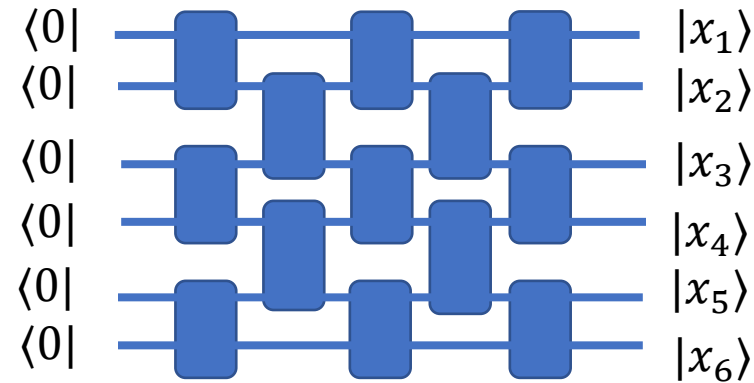
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Tensor network contraction methods:

[Markov Shi 2005][Pednault et al. 2017][Boixo et al. 2017][Li et al. 2018][Chen et al. 2018]

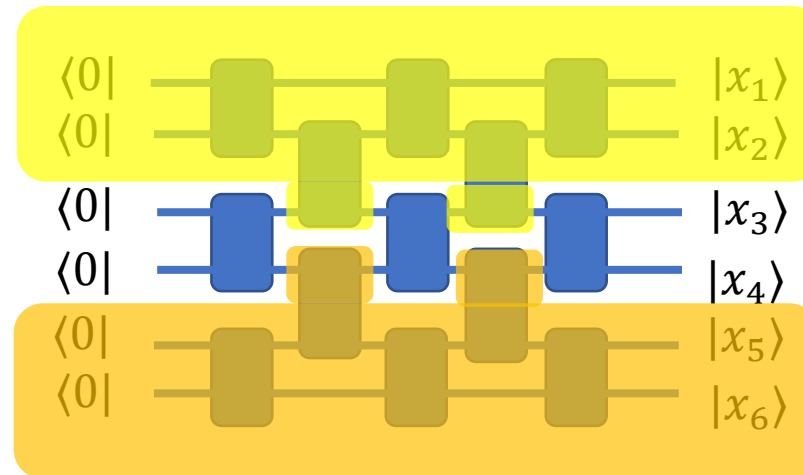
Tensor network contraction methods

Simulation algorithm idea: view circuit as a tensor network and choose contraction ordering to minimize memory/runtime.



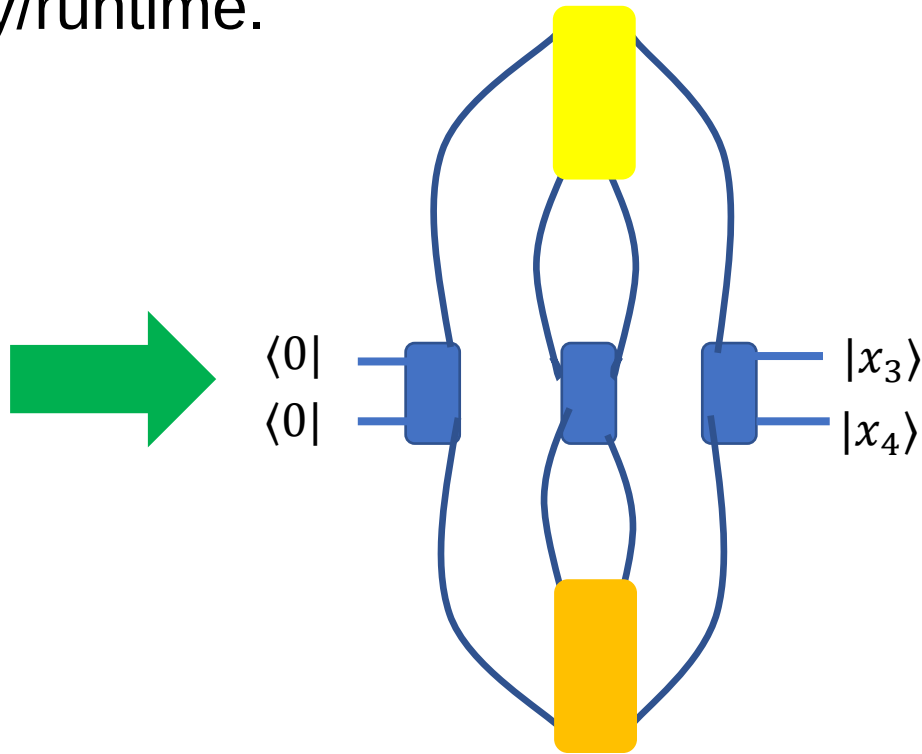
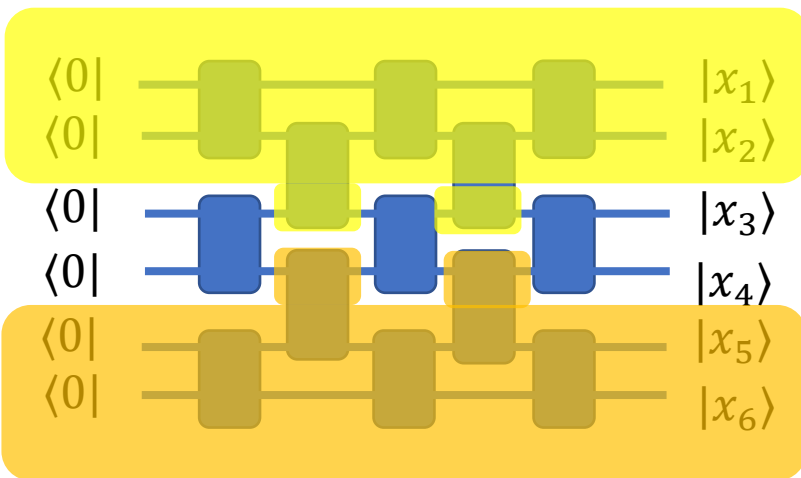
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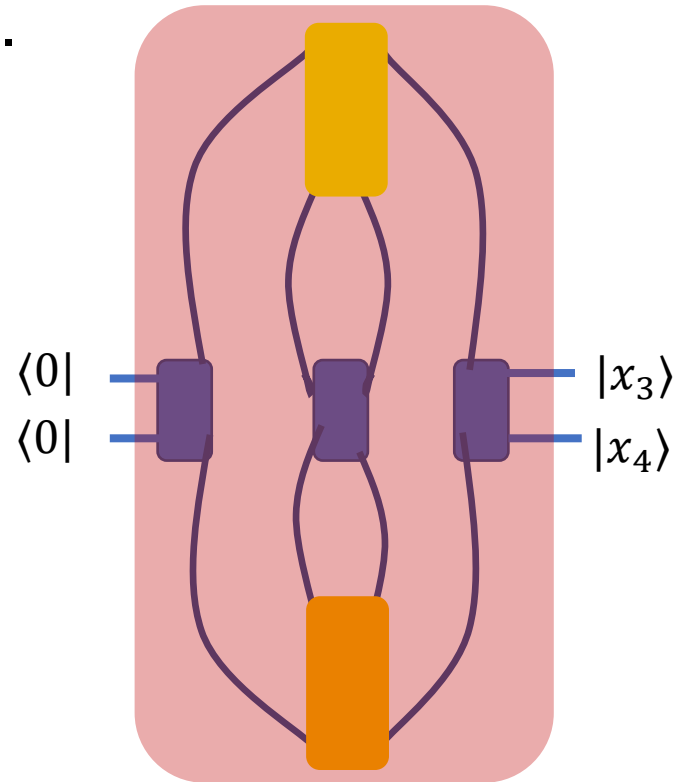
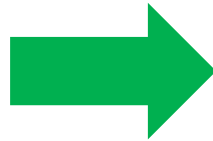
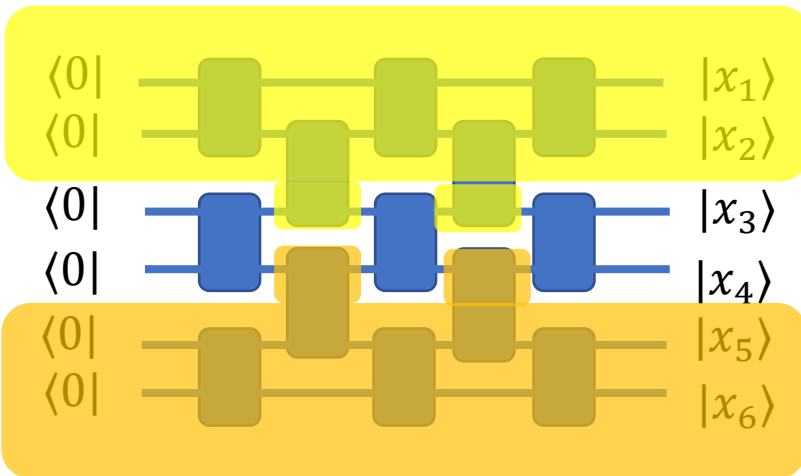
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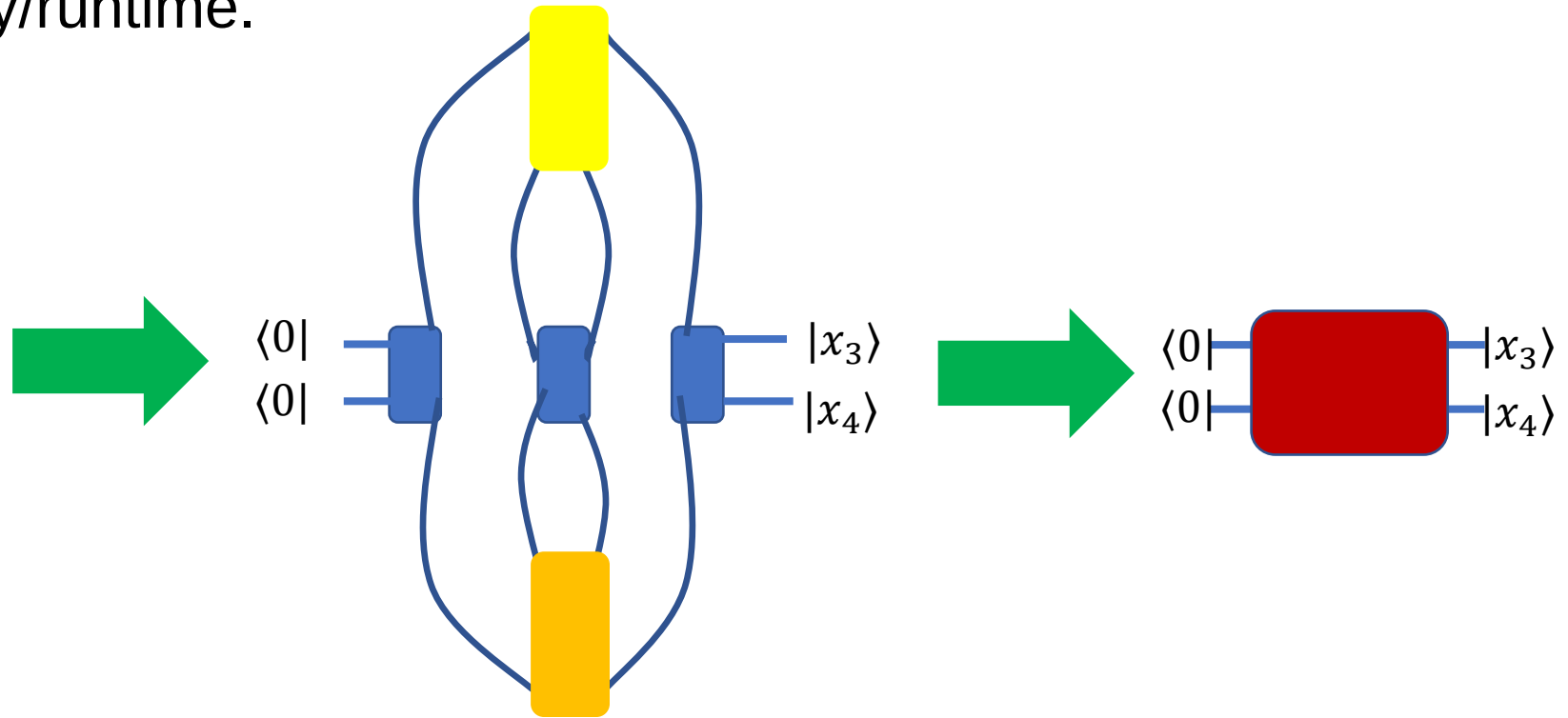
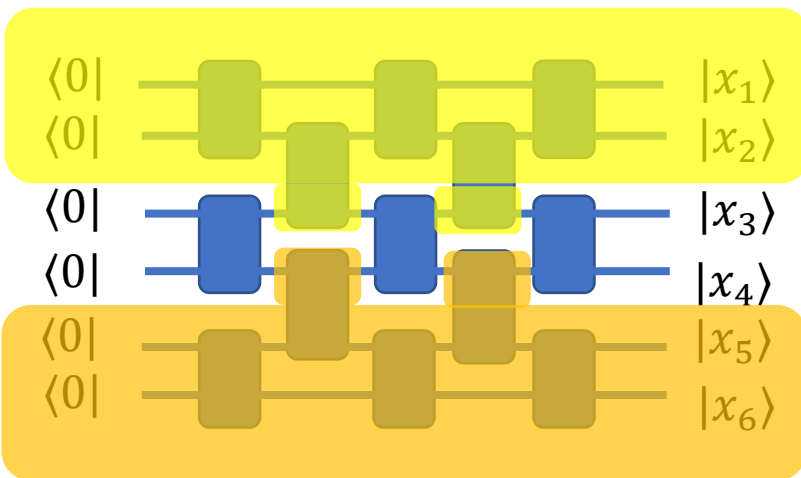
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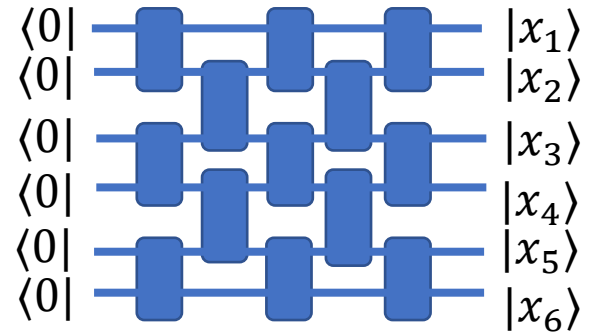


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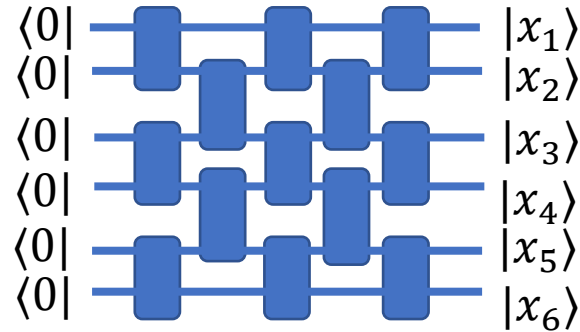


Tensor network contraction methods



The runtime of tensor network contraction algorithms depend on the connectivity of the circuit and is **insensitive to the entries of the gates** that appear.

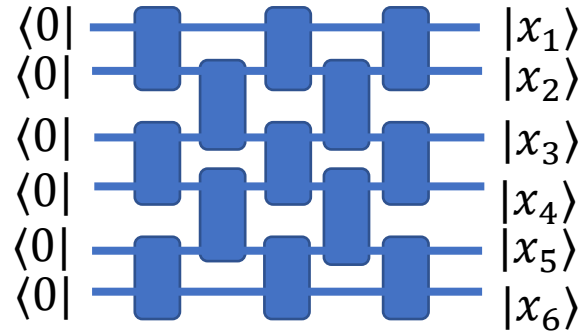
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This talk is about a different kind of simulation algorithm...

Stabilizer rank simulators for Clifford+T circuits

Clifford circuits

The **Clifford group** is generated by gates

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

$$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Gottesman-Knill Theorem [Gottesman 1997]

Quantum circuits composed only of Clifford gates can be efficiently (weakly/strongly) simulated on a classical computer.

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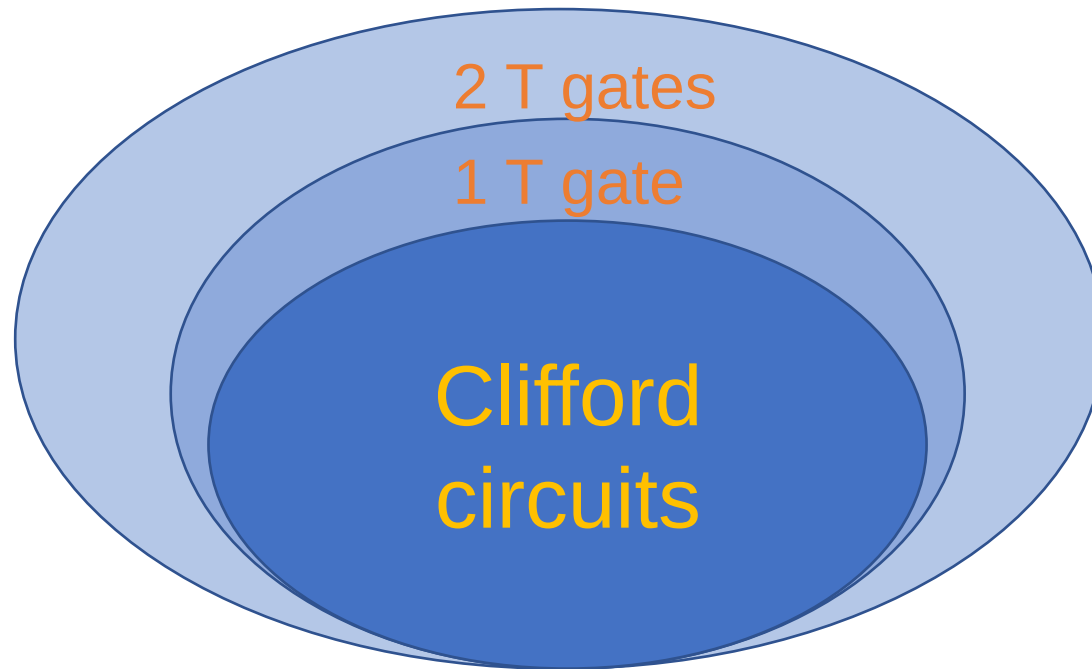
Gottesman-Knill Theorem [Gottesman 1997]

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Can we extend Gottesman-Knill to circuits with a few non-Clifford gates?...

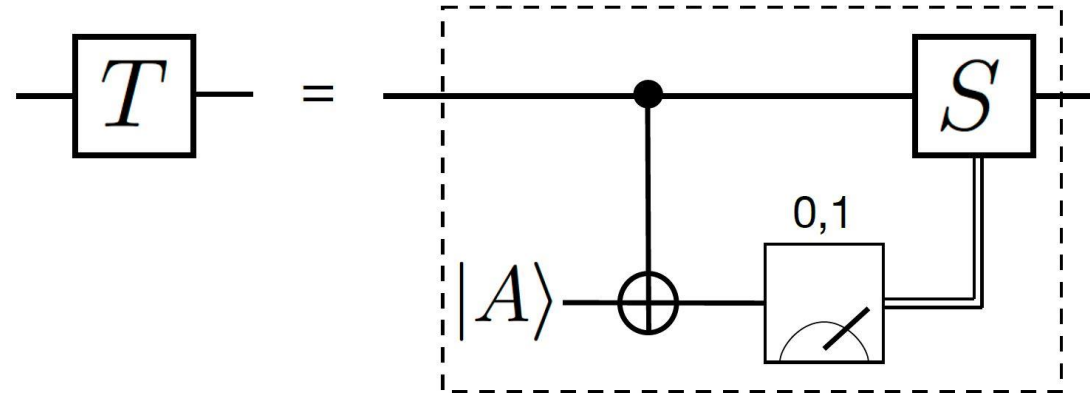
Outward from the Cliffords

Clifford+T gate set $\{H, S, CNOT, T\}$ $T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$



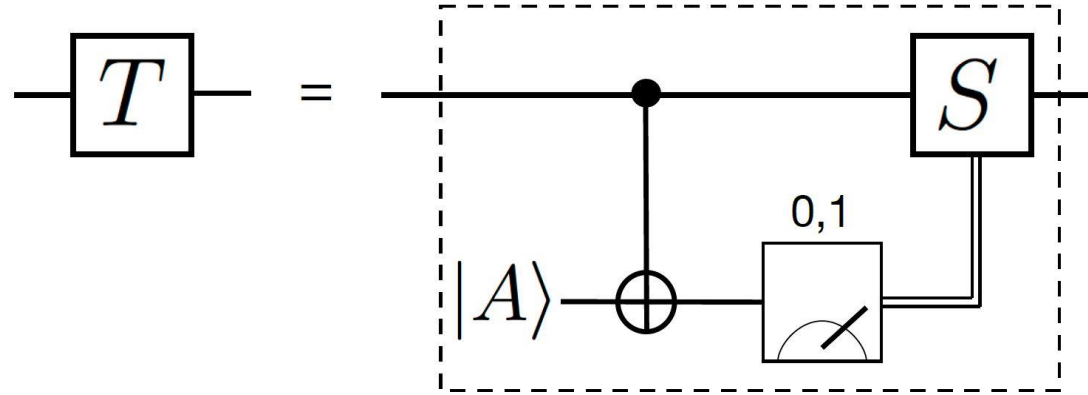
What is the classical simulation cost of a circuit with m **T gates**?

Gadgetized Clifford+T circuit



T gate gadget
[Zhou Leung Chuang 2000]

Gadgetized Clifford+T circuit



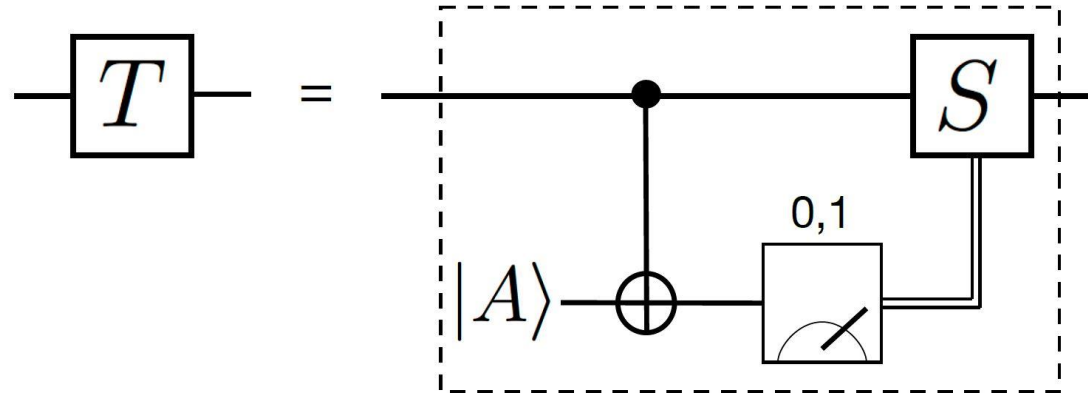
$$|A\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi/4} |1\rangle)$$

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Magic state

Gadgetized Clifford+T circuit



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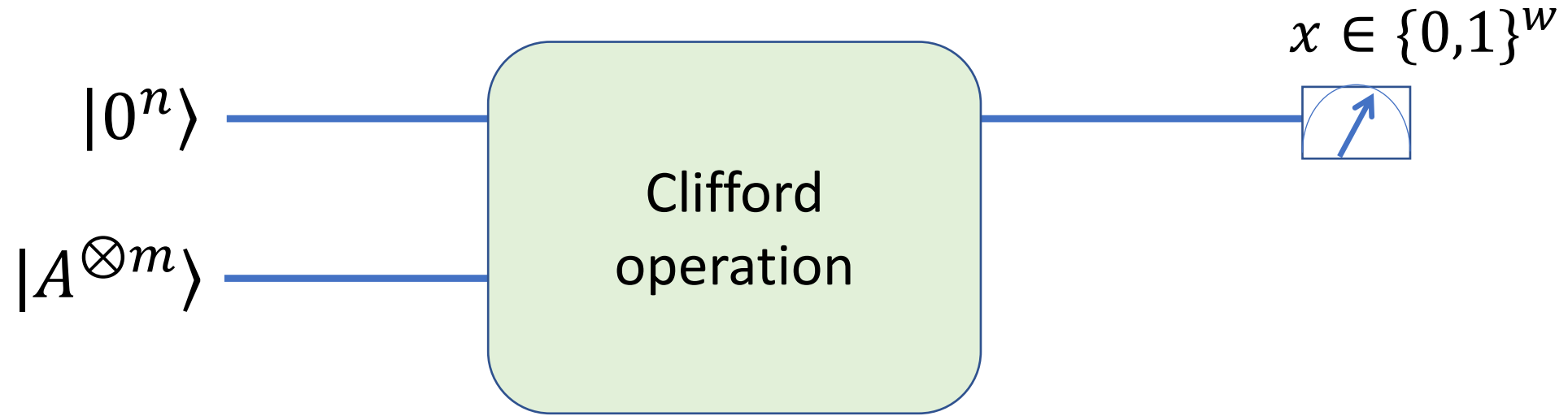
Magic state

We can **gadgetize** any Clifford+T circuit by replacing all T gates with the above gadget.

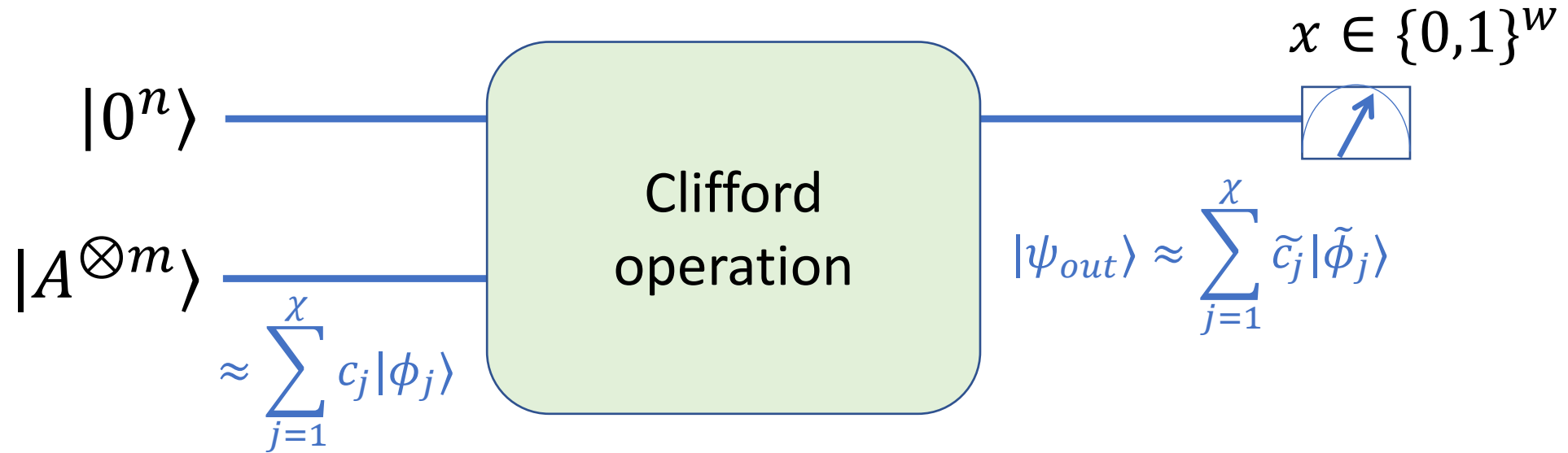
This gives an adaptive Clifford circuit with input state

$$|0^n\rangle |A^{\otimes m}\rangle$$

Simulating the gadgetized circuit [Bravyi Smith Smolin 2015]



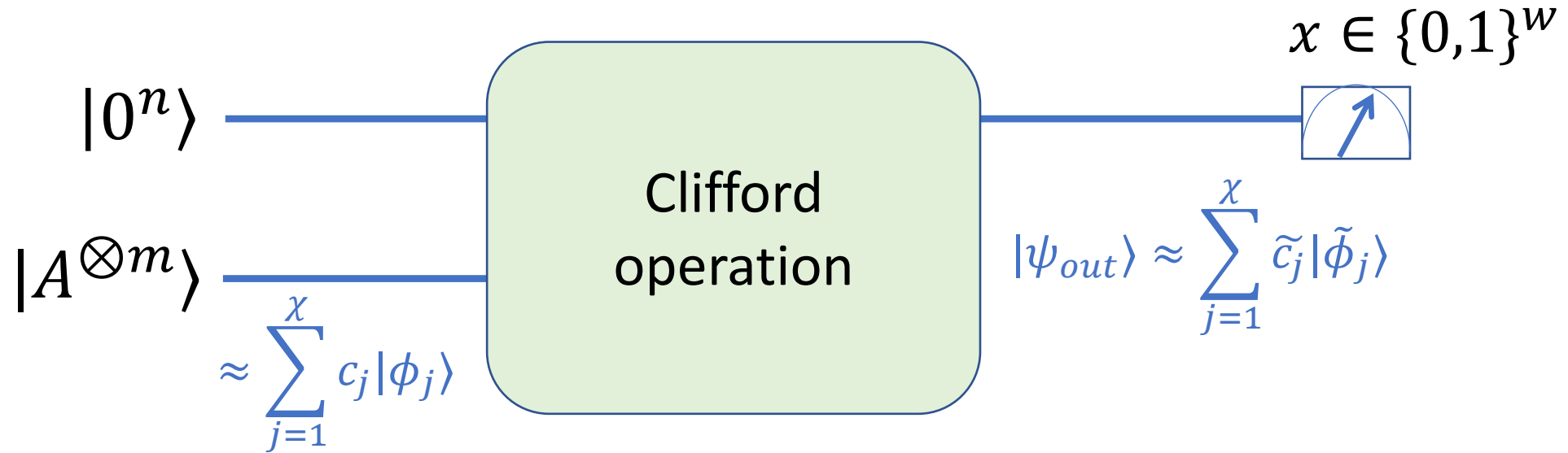
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Simulation strategy (roughly):

1. Approximate input magic state as superposition of $\chi \ll 2^n$ stabilizer states
2. Apply Clifford operation
3. Simulate final measurement

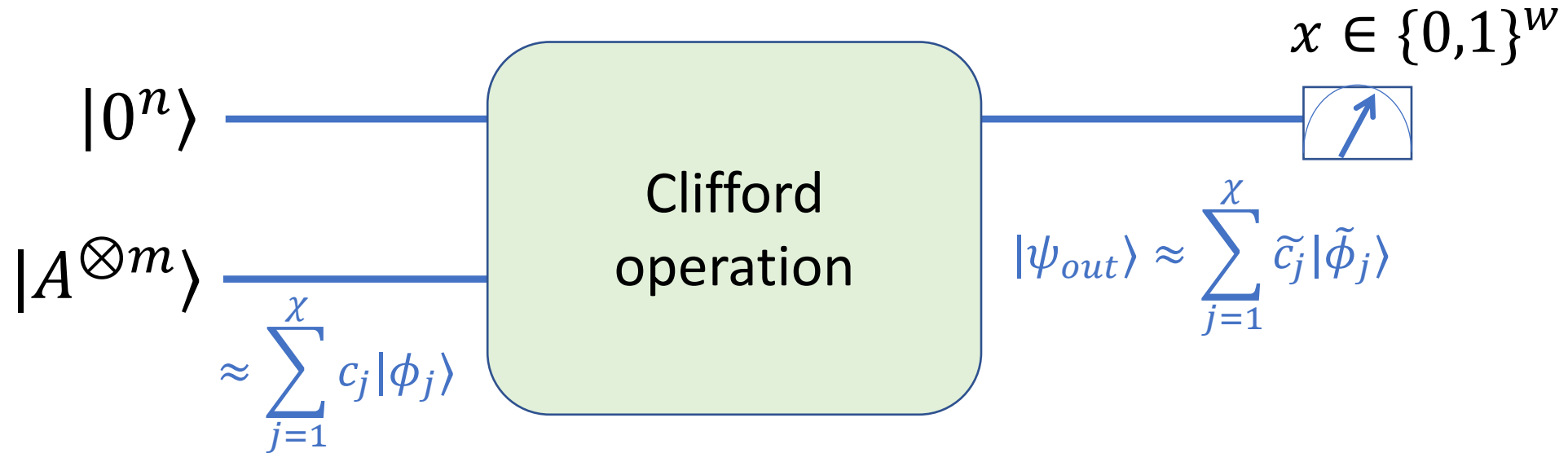
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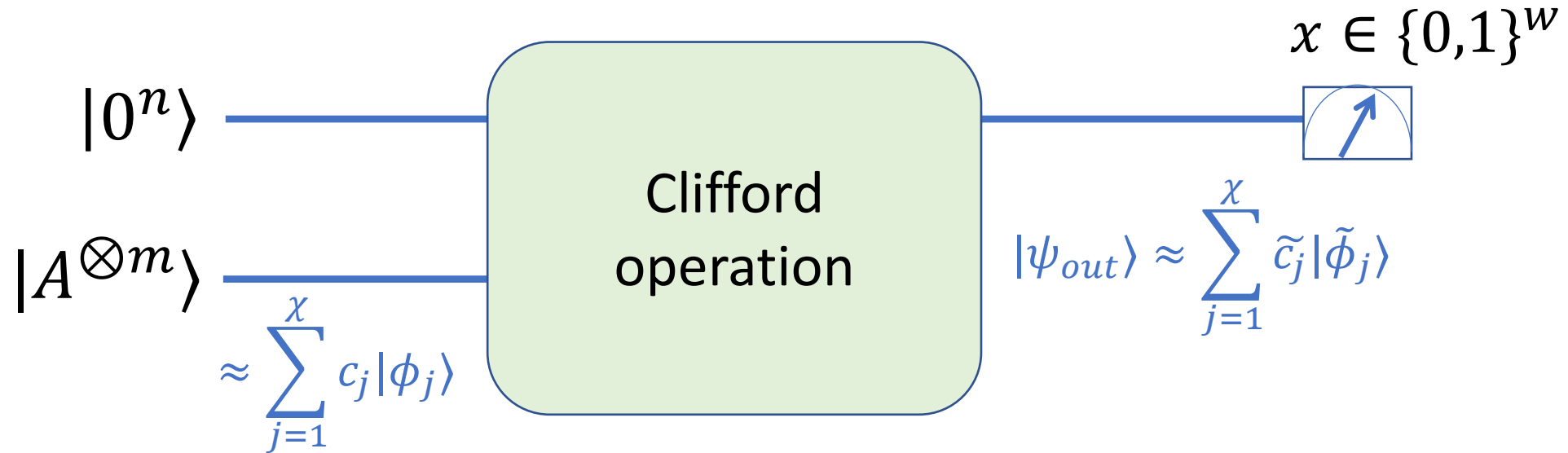


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Runtime:

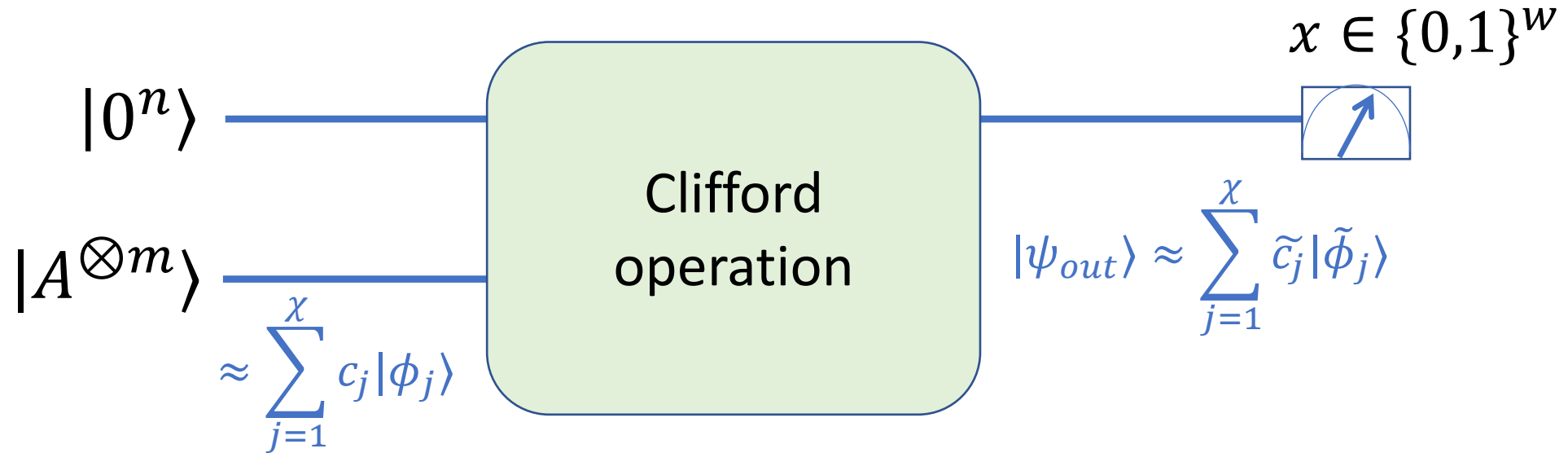
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Involves computation of marginals $\langle \psi_{out} || x \rangle \langle x | \otimes I | \psi_{out} \rangle$ $\chi^2 \cdot \text{poly}(m, n)$

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“norm estimation algorithm”
(weak or ϵ -strong sim.)
[Bravyi DG 2016]

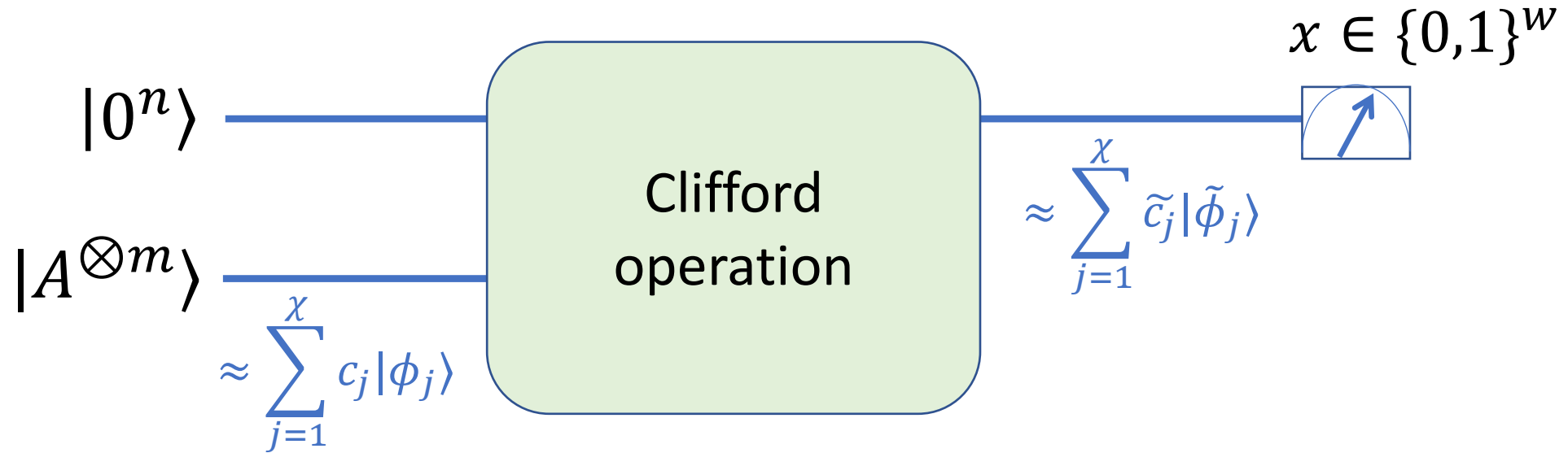
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What kind of approximation is needed? How big is χ ?...

Stabilizer decompositions

Stabilizer rank $\chi(\psi)$ is the minimum r such that

$$|\psi\rangle = \sum_{j=1}^r c_j |\phi_j\rangle$$

← Stabilizer states

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Stabilizer states



Approximate stabilizer rank $\chi_\delta(\psi)$ is the minimum r such that

$$\left\| |\psi\rangle - \sum_{j=1}^r c_j |\phi_j\rangle \right\| \leq \delta$$

Remark on stabilizer decompositions

Recall that a stabilizer state can be parameterized as

$$|\phi\rangle \propto \sum_{x \in V} (-1)^{q(x)} i^{\ell(x)} |x\rangle$$

V : affine subspace of $\mathbb{F}_2^n$

q : quadratic function $q(x) = x^T B x \pmod{2}$

ℓ : linear function $\ell(x) = d^T x \pmod{2}$

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The decomposition of a given state into the overcomplete basis of stabilizer states generalizes the decomposition of a Boolean function into quadratic Boolean functions (**quadratic Fourier analysis**)

Stabilizer rank simulators

Classical simulation cost scales with stabilizer rank of magic state input

For circuits with **n qubits, g total gates, m T gates:**

[Bravyi Smith Smolin 2015]	$\left(\chi(A^{\otimes m})\right)^2 \text{poly}(n, g)$	Strong simulation
[Bravyi, DG 2016]	$\chi(A^{\otimes m}) \text{poly}(n, g)$	ϵ-Strong simulation
[Bravyi, DG 2016]	$\chi_\delta(A^{\otimes m}) \text{poly}(n, g)$	Weak simulation

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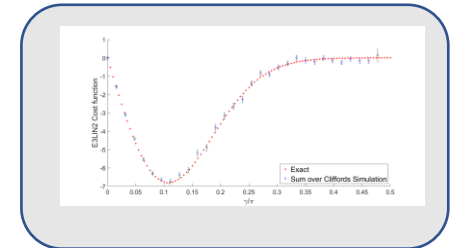
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Weak simulation

Remarks:

Techniques can be extended to circuits with other non-Clifford gates

[Bravyi Browne Calpin Campbell DG Howard 18]



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Remarks:

Up to polynomial factors, classical simulation has **linear scaling** with stabilizer rank.

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Remarks:

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The best unconditional lower bound is $\chi(A^{\otimes m}) = \Omega(m)$ [Peleg, Shpilka, Volk 2021]

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Remarks:

Rest of this talk: upper bounds...

Upper bounds on stabilizer rank of magic states

Upper bound on approximate SR [Bravyi DG 2016]

$$\chi_{\delta}(A^{\otimes m}) \leq O\left(\frac{1}{\delta^2} 2^{\alpha m}\right)$$

$$\alpha = -2\log_2(\cos \pi/8) \approx 0.228 \dots$$

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Proof sketch $|H\rangle = \frac{1}{2\cos(\frac{\pi}{8})} (|0\rangle + |+\rangle)$ **Clifford equivalent to $|A\rangle$**

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Clifford equivalent to $|A\rangle$

$$|H\rangle^{\otimes m} = \frac{1}{\left(2\cos\left(\frac{\pi}{8}\right)\right)^m} \sum_{x \in \{0,1\}^m} |\hat{x}\rangle$$



$$|\hat{0}\rangle = |0\rangle$$

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$\left. \begin{array}{l} |\hat{0}\rangle = |0\rangle \\ |\hat{1}\rangle = |+\rangle \end{array} \right\}$

A low stabilizer rank approximation is given by restricting the sum to $x \in L \subseteq \{0,1\}^m$ where L is a randomly chosen linear subspace of size $|L| \approx \delta^{-2} 2^{\alpha m}$

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Scaling with δ was improved in [Seddon Regula Pashayan Ouyang Campbell 20]

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Can show it is optimal under certain restrictions on the stabilizer states appearing in the decomposition.

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Is the scaling with m optimal?

Can show it is optimal under certain restrictions on the stabilizer states appearing in the decomposition.

Can also show the corresponding stabilizer decomposition has minimal 1-norm of the coefficients in the decomposition

(“stabilizer extent”) [Bravyi Browne Calpin Campbell DG Howard 2018]

In contrast, the known upper bounds on the **exact** stabilizer rank of magic states seem unlikely to be optimal...

Upper bounds on exact stabilizer rank

All previously known upper bounds follow a similar strategy

1. Upper bound SR of a constant number of magic states

Upper bound $\chi(A^{\otimes c})$ for some small constant number of magic states c

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1. Upper bound SR of a constant number of magic states

Upper bound $\chi(A^{\otimes c})$ for some small constant number of magic states c

2. Use (trivial) submultiplicativity property $\chi(\phi \otimes \psi) \leq \chi(\phi)\chi(\psi)$

$$\chi(A^{\otimes m}) \leq \chi(A^{\otimes c})^{\frac{m}{c}}$$

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This step is not as easy as it sounds!

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Upper bound $\chi(A^{\otimes c})$ for some small constant number of magic states c

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expect that the scaling in Theorem 4 can be improved by computing χ_n for larger values of n . In Appendix B we describe a heuristic algorithm for computing low-rank decompositions of $|H^{\otimes n}\rangle$ into stabilizer states which yields the following upper bounds:

n	2	3	4	5	6
$\chi_n \leq$	2	3	4	6	7

We believe that these upper bounds are tight. A lower bound $\chi_n \geq \Omega(n^{1/2})$ is proved in Appendix C.

[From Bravyi Smith Smolin 2015]

Upper bounds on exact stabilizer rank

All previously known upper bounds follow a similar strategy

1. Upper bound SR of a constant number of magic states

Upper bound $\chi(A^{\otimes c})$ for some small constant number of magic states c

This step is not as easy as it sounds!

expect that the scaling in Theorem 4 can be improved by computing χ_n for larger values of n . In Appendix B we describe a heuristic algorithm for computing low-rank decompositions of $|H^{\otimes n}\rangle$ into stabilizer states which yields the following upper bounds:

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of 6-qubit stabilizer states: 315057600

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of size-7 subsets of
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[Kocia 20] shows (by inspection?) $\chi(A^{\otimes 12}) \leq 47$

Upper bounds on exact stabilizer rank

2. Use (trivial) submultiplicativity property $\chi(\phi \otimes \psi) \leq \chi(\phi)\chi(\psi)$

Using the bound $\chi(A^{\otimes 6}) \leq 7$ from [Bravyi Smith Smolin 15] gives

$$\chi(A^{\otimes m}) \leq 2^{\alpha m} \quad \alpha = \frac{\log_2 7}{6} \approx 0.4679 \dots$$

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Using the bound $\chi(A^{\otimes 12}) \leq 47$ from [Kocia 20] gives

$$\chi(A^{\otimes m}) \leq 2^{\alpha m} \quad \alpha = \frac{\log_2 47}{12} \approx 0.4629 \dots$$

In [Qassim Pashayan DG 21] we improve on this bound using a different strategy...

Magic cat states

Consider the magic cat state



Image source: Etsy

$$|\text{Cat}_m\rangle = \frac{1}{\sqrt{2}} \left(|A\rangle^{\otimes m} + |A^\perp\rangle^{\otimes m} \right)$$

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Stabilizer projector—does not increase SR

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$$\frac{1}{2} \cdot \chi(A^{\otimes m}) \leq \chi(\text{Cat}_m) \leq \chi(A^{\otimes m})$$

$$|A\rangle^{\otimes m} = (|A\rangle\langle A| \otimes I) |\text{Cat}_m\rangle$$

This single-qubit operator can be written as a sum of two Cliffords, each of which does not increase SR

Magic cat states

Consider the magic cat state



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Stabilizer rank of magic cat has same asymptotic scaling with m

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Stabilizer rank of magic cat has same asymptotic scaling with m

Stabilizer decompositions of magic cats with small m are easier to find...

Stabilizer rank of small magic cats

$$|\text{Cat}_1\rangle = \frac{1}{\sqrt{2}} (|A\rangle + |A^\perp\rangle) = |0\rangle$$

Cat1 is a stabilizer state

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Cat2 is also a stabilizer state

$$\chi(\text{Cat}_2) = 1$$

Stabilizer rank of small magic cats

The largest magic cat state for which we can exactly compute the stabilizer rank is

$$\chi(\text{Cat}_6) = 3$$

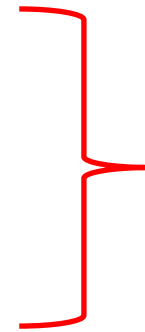
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$$|\text{Cat}_6\rangle = \frac{1}{2}|\phi_1\rangle + \frac{e^{\frac{3i\pi}{4}}}{\sqrt{2}}(|\phi_2\rangle + i|\phi_3\rangle)$$



$$|\phi_1\rangle = \frac{1}{\sqrt{2}}(|0^6\rangle - i|1^6\rangle)$$

$$|\phi_2\rangle = 2^{-5/2} \sum_{|x| \text{ even}} |x\rangle$$

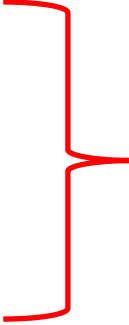
$$|\phi_3\rangle = \prod_{i < j} CZ_{ij} |\phi_2\rangle$$

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(We establish a matching lower bound by showing that no state with stabilizer rank 2 can have the same set of Pauli expectation values)

Low hanging fruit

$$\chi(A^m) \leq 2\chi(\text{Cat}_m)$$

$$\chi(\text{Cat}_6) = 3 \quad \longrightarrow \quad \chi(A^{\otimes 6}) \leq 6$$

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Falsifies conjecture that

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Submultiplicativity

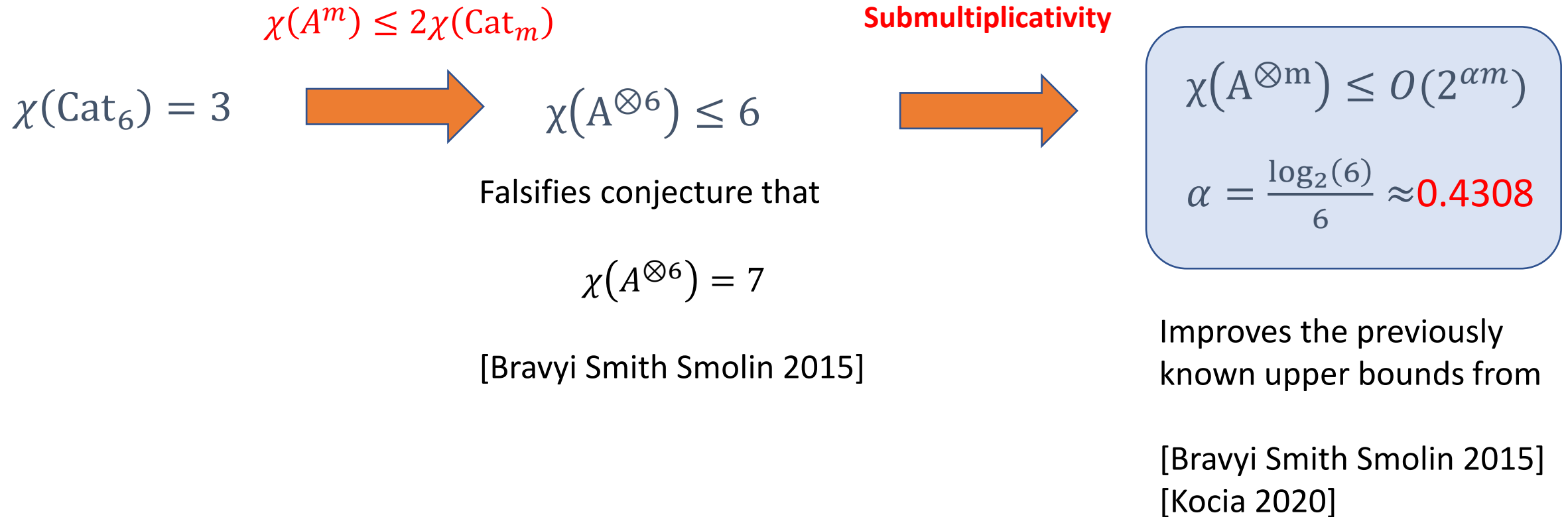
$$\chi(A^{\otimes m}) \leq O(2^{\alpha m})$$

$$\alpha = \frac{\log_2(6)}{6} \approx \mathbf{0.4308}$$

Improves the previously
known upper bounds from

[Bravyi Smith Smolin 2015]
[Kocia 2020]

Low hanging fruit



We can construct better stabilizer decompositions of $|A\rangle^{\otimes m}$ by using magic cats in a different way...

Stabilizer rank of large magic cats

Can we build big magic cat states out of small ones?



Image source: Wikipedia

Stabilizer rank of large magic cats

Can we build big magic cat states out of small ones?

$$|\text{Cat}_m\rangle |\text{Cat}_m\rangle = \frac{1}{2} \left(|A\rangle^m + |A^\perp\rangle^m \right) \left(|A\rangle^m + |A^\perp\rangle^m \right)$$

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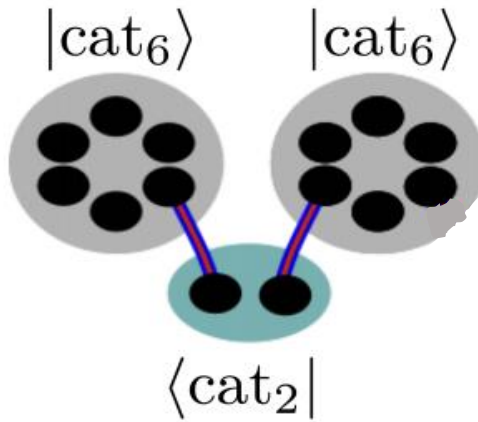
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$$\chi(\text{Cat}_{2m-2}) \leq (\chi(\text{Cat}_m))^2$$

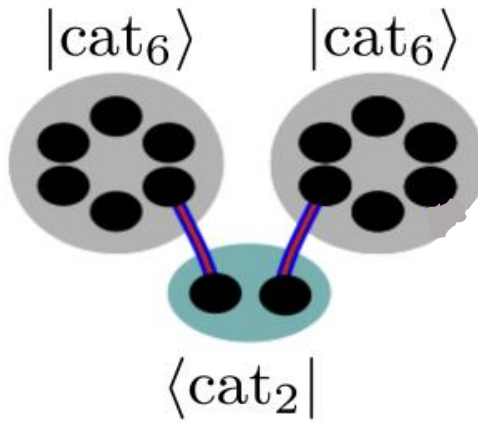
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So for example, contracting two copies of Cat6 with one copy of Cat2 in this way gives



Stabilizer rank of large magic cats

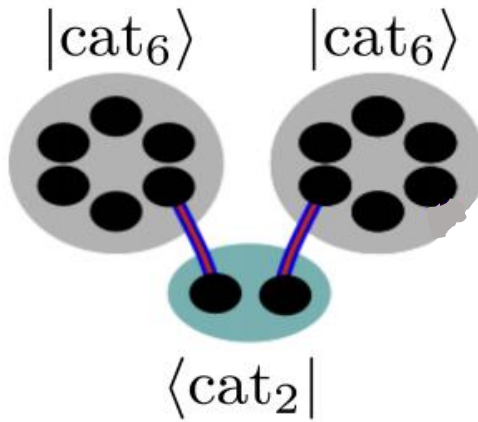
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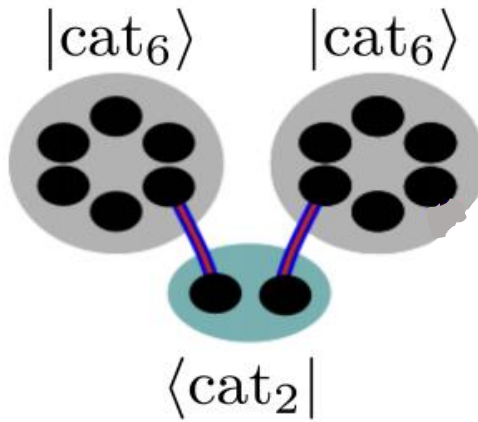


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$$\chi(A^{10}) \leq 18$$

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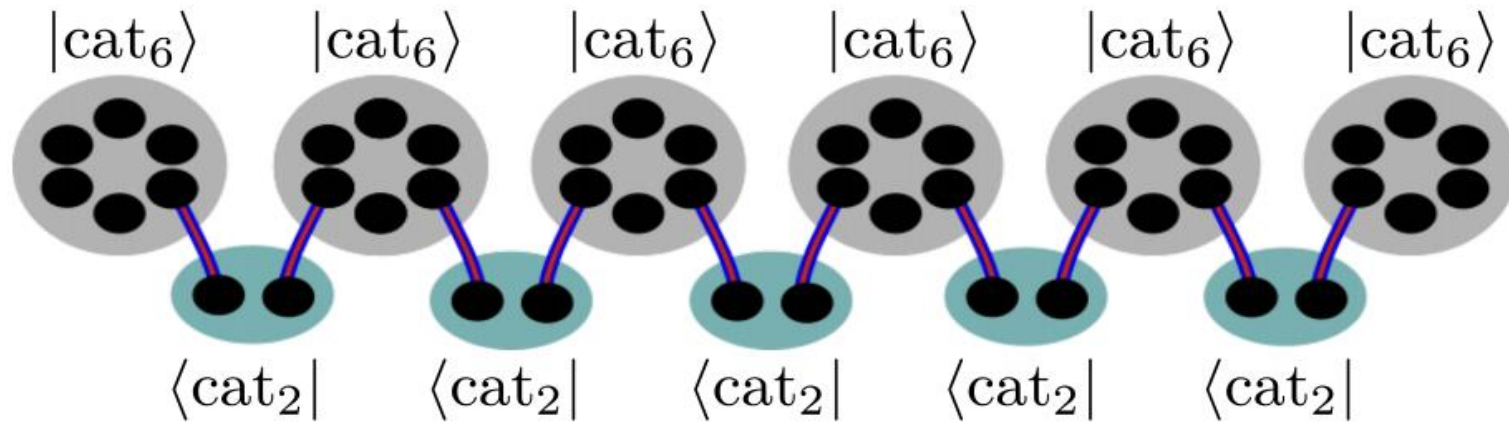
$$\chi(A^{10}) \leq 18$$

$$\chi(A^m) \leq 2^{\alpha m}$$

$$\alpha \leq \frac{\log_2(18)}{10} = 0.41699 \dots$$

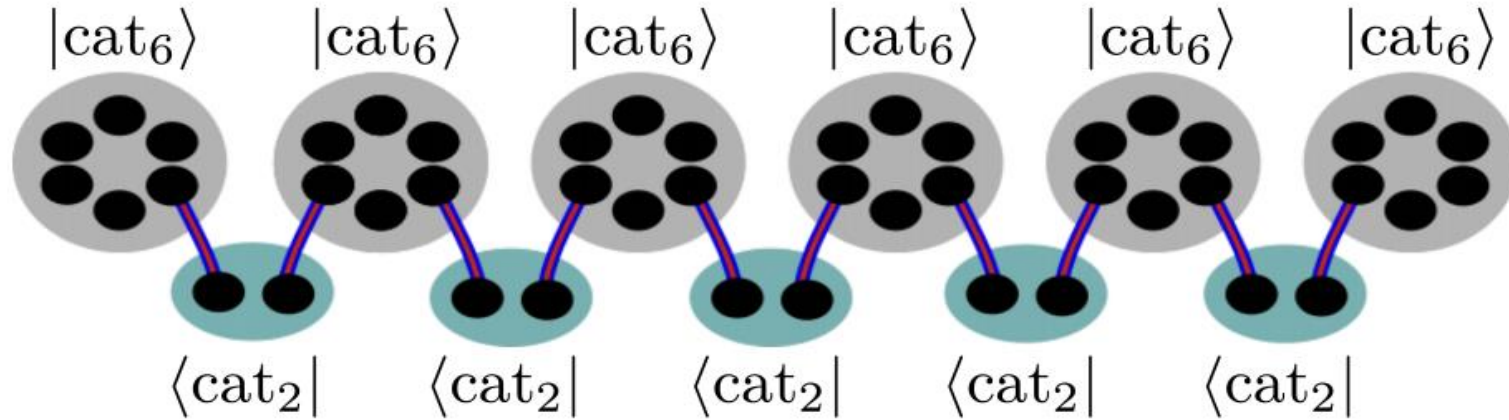
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An even better strategy is to contract many copies of Cat6 arranged in a chain



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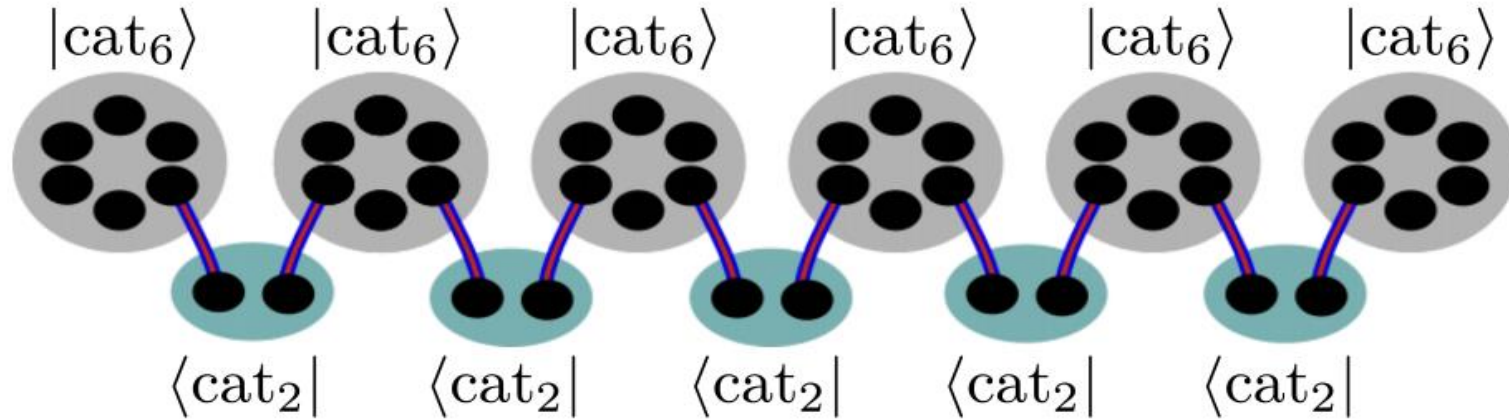
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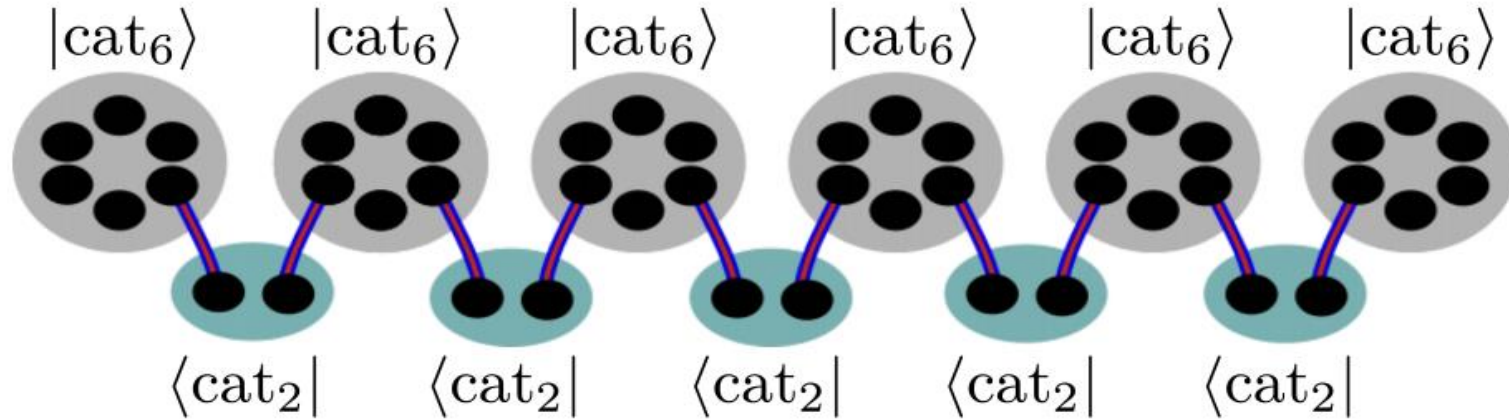
Taking L to be large and setting $m = 4L - 2$ we get

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$$\alpha = \frac{\log_2(3)}{4} = \mathbf{0.3962 \dots}$$

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Upper bounds (up to polynomial factors) the runtime of ϵ -strong simulation of Clifford+T circuits as a function of the number of T gates m

There is still a gap between this scaling and that of approx. SR (weak simulation cost) where the best upper bound has $\alpha \approx 0.228 \dots$

Extensions

Stabilizer rank of symmetric states: Using a similar technique we upper bound the stabilizer rank of many copies of any given equatorial single-qubit state. Using the fact that such states span the symmetric subspace we get

$$\chi(\psi^{\otimes m}) \leq O(2^{m/2})$$



Any single-qubit state

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Magic code states: We can generalize the magic cat state to a family of states which are superpositions over the codewords of a linear code $L \subseteq \{0,1\}^m$

$$|L\rangle = \sum_{x \in L} |\hat{x}\rangle \quad \left. \begin{array}{l} |\hat{0}\rangle = |A\rangle \\ |\hat{1}\rangle = |A^\perp\rangle \end{array} \right\}$$

Cat state is the special case where L is the repetition code

We show how an upper bound on the stabilizer rank of magic states follows from an upper bound on $\chi(|L\rangle)$ for any code with dimension $k \leq m/2$

Open questions

Describe any nontrivial criterion for certifying that a given state has stabilizer rank at least k

Establish superpolynomial lower bound on $\chi(A^m)$

Improve upper bounds on $\chi(A^m)$

Applications to classical counting problems?

m	1	2	3	4	5	6	7	8
$\chi(A^{\otimes m})$	2	2	3	≤ 4	≤ 6	≤ 6	≤ 12	≤ 12
$\chi(\text{Cat}_m)$	1	1	2	2	3	3	≤ 6	≤ 6

Thanks!